

KAPIL KUMAR MEENA

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Education

Indian Institute of Technology Kharagpur <i>PhD in Infrastructure Design & Management</i> Focus: <i>Travel Behaviour, Mobility, and Air Quality</i> Advisor: Prof. Arkopal Goswami	West Bengal, India 2022–Present
Indian Institute of Technology Roorkee <i>M.Tech in Transportation Engineering</i> Thesis: <i>Commuter Exposure to Air Pollution</i> Advisor: Prof. Amit Agarwal	Uttarakhand, India 2019–2021
Rajasthan Technical University <i>B.Tech in Civil Engineering (Honors)</i> Graduated with Honors Top 5% of class	Kota, India 2015–2019

Experience

World Resources Institute (WRI) India <i>Research Consultant (Intern)</i>	Remote Nov 2021 – Mar 2022
○ Conducted comparative analysis of inhaled pollution dose for commuters inside electric vs. ICE vehicles. ○ Developed policy recommendations for reducing commuter exposure in Tier-2 cities. ○ Advisor: Aparna Vijaykumar, Program Manager - Electric Mobility.	
IIT Kharagpur & IIT Roorkee <i>Teaching Assistant</i>	India 2019–2023
○ Courses: Engineering Drawing (CE13001), Multimodal Urban Transport (NPTEL), Intersection Design (CEN 662). ○ Mentored junior students in laboratory sessions and software workshops (VISSIM, QGIS).	

Key Projects & Tools

DRUM Web App: [Developed the *Dynamic Route Planning for Urban Mobility* application.](#) A web tool offering sustainable, pollution-aware route options. **Featured in The Hindu** (2025).

TUTEM Project: [Researcher for "Technologies for Urban Transit to Enhance Mobility" \(ADB-Funded\).](#) Focusing on first/last-mile connectivity and safety for vulnerable road users.

Air Quality Dashboard: [Built an interactive Streamlit application](#) visualizing AQI data across major Indian cities using real-time API integration.

PM2.5 Forecasting: [Machine Learning model for predicting next-day PM2.5 concentrations.](#)

MaaS Demand Study: Led a demand forecasting study for Mobility-as-a-Service in India using discrete choice experiments (2022).

Publications

Journal Articles.....

2026: Meena, K. K., A.K. Goswami. "Not all travellers think alike: Segmenting travel behaviour under air pollution exposure using a hybrid latent class and discrete choice approach." *Transportation Research Record* (TRR). (In 2nd Review stage)

2025: Meena, K. K., A.K. Singh, A.K. Goswami. "Dynamic Route Planning for Urban Green Mobility: Development of a Web Application Offering Sustainable Route Options to Commuters." *Transportation Research Record*. (Published May 2025) [\[DOI\]](#)

- 2025:** B.S. Manoj, **Meena, K. K.**, H. Panchal, G. Sharma, A.K. Goswami. "Modeling bicycle choice behavior and its potential health impact: Case of first/last mile access to suburban rail." *International Journal of Sustainable Transportation*, pp. 1-21. (Published Oct 2025) [\[Link\]](#)
- 2025:** B.S. Manoj, **Meena, K. K.**, A.K. Goswami. "A prioritization framework to identify key attributes of transit-oriented development (TOD) using multi-criteria decision-making (MCDM) approach: an Indian context." *Sustainable Transport and Livability*, vol. 2. (Published Jul 2025) [\[DOI\]](#)
- 2024:** **Meena, K. K.**, A.K. Goswami. "A review of air pollution exposure impacts on travel behaviour and way forward." *Transport Policy*, Elsevier. (Published Aug 2024) [\[DOI\]](#)
- 2024:** **Meena, K. K.**, D. Bairwa, A. Agarwal. "A machine learning approach for unraveling the influence of air quality awareness on travel behavior." *Decision Analytics Journal*, vol. 11, 100459. Elsevier. (Published Jun 2024) [\[DOI\]](#)
- 2023:** **Meena, K. K.**, V. Singh, A. Agarwal. "Perception of commuters towards air quality in Delhi." *Journal of Transport & Health*, vol. 31, 101643. Elsevier. (Published Jun 2023) [\[DOI\]](#)
- 2021:** V. Singh, **Meena, K. K.**, A. Agarwal. "Travellers' exposure to air pollution: A systematic review and future directions." *Urban Climate*, vol. 38, 100901. Elsevier. (Published Jul 2021) [\[DOI\]](#)
- Conference Proceedings (Refereed)**.....
- 2026:** **Meena, K. K.**, A.K. Goswami. "Not all travellers think alike: Segmenting travel behaviour under air pollution exposure using a hybrid latent class and discrete choice approach." *Transportation Research Board (TRB) 2026 Annual Meeting*, Washington, DC. **(Accepted)**
- 2026:** C. Gupta, A. Amitabh, **Meena, K. K.**, A.K. Goswami. "A Hybrid Geostatistical and Deep Learning Framework for Urban Pollutant Concentration Prediction from Sparse Data." *18th International Conference on COMMunication Systems & NETworkS (COMSNETS)*. (Accepted)
- 2026:** A. Singh, **Meena, K. K.**, G. Sharma, A.K. Goswami, S. Mishra. "Developing an integrated walkability score using image-based feature extraction and user preferences." *Transportation Research Board (TRB) 2026 Annual Meeting*, Washington, DC. (Submitted)
- 2025:** **Meena, K. K.**, A.K. Singh, A.K. Goswami. "Dynamic route planning for urban green mobility." *7th International Conference of Transportation Research Group of India (CTRG-2023)*, SVNIT Surat, India.
- 2025:** A. Sumbhate, **Meena, K. K.**, A.K. Goswami. "Assessing the air pollution exposure to school children in different modes of transport while commuting to school: A case of Kharagpur, India." *Proceedings of the Eastern Asia Society for Transportation Studies (EASTS)*. **(Accepted)**
- 2025:** B.S. Manoj, **Meena, K. K.**, A.K. Goswami. "A prioritization framework to identify key attributes of transit-oriented development (TOD)." *1st World Symposium on Sustainable Transport and Livability (WSSTL-2025)*, IISc Bengaluru.
- 2025:** R. Kodukulla, **Meena, K. K.**, G. Sharma, A.K. Goswami. "Accessibility assessment of urban public transit to key facilities through spatial analysis a case study of Delhi." *Transportation Infrastructure Projects: Conception to Execution (TIPCE)*, IIT Roorkee. **(Accepted)**
- 2025:** S. Dasgupta, **Meena, K. K.**, D. Majumdar, A.K. Goswami. "Air pollution exposure among Kolkata's auto-rickshaw drivers: Pm variability, health risks, and predictive modeling." *Energies, AEEE India*. **(Accepted)**
- 2024:** **Meena, K. K.**, R. Taneja, A. Agarwal. "Impact of air pollution on informed decision-making for choice of a travel mode." *16th International Conference on COMMunication Systems & NETworkS (COMSNETS)*. IEEE. [\[DOI\]](#)
- 2024:** A. Sumbhate, **Meena, K. K.**, A.K. Goswami. "Breathable modes to school: Assessing the air pollution exposure of travel choices for school children in urban environments." *52nd Urban Affairs Association (UAA) Annual Meeting*, Nashville, USA.
- 2024:** P. Mohanty, **Meena, K. K.**, A.K. Goswami. "Analysing user behaviour along dedicated bicycle facilities in an urban environment." *52nd Urban Affairs Association (UAA) Annual Meeting*, Nashville, USA.
- 2024:** B.S. Manoj, **Meena, K. K.**, H. Panchal, G. Sharma, A.K. Goswami. "Assessing the willingness to bicycle for the first mile to the Mumbai suburban rail." *17th International Association for Travel Behaviour Research (IATBR)*, Vienna, Austria.

2023: Meena, K. K., A.K. Goswami. "A review of air pollution exposure impacts on travel behaviour and way forward." *16th World Conference on Transport Research (WCTR)*, Montréal, Canada.

2022: Meena, K. K., R. Kumar, A.K. Goswami. "On-road pollution exposure in multiple transport micro-environments: A case study of tier-2 and tier-3 cities in india." *14th TPMDC*, IIT Bombay, India.

Media Features.....

2025: Ashmita Gupta. "IIT-KGP app helps commuters pick 'greener' routes on the road." *The Hindu*, June 2025. Feature article on DRUM web app developed by IIT Kharagpur researchers, including **Meena, K. K.**

Patent.....

2025 (In Process): Meena, K. K. and Arkopal Kishore Goswami. "Personalized dynamic route planning system for sustainable urban mobility."

Technical Expertise

- **Programming & Web:** Python (Pandas, Scikit-learn), R, JavaScript, ReactJS, Streamlit.
- **Data Analysis & GIS:** Tableau, QGIS, SPSS, NLOGIT, Biogeme (Choice Modeling), MiniTab.
- **Databases & Tools:** MySQL, MongoDB, LaTeX, JabRef, Git/GitHub.

Leadership & Service

Session Chair: Technical Session Chair, Eastern Asia Society for Transportation Studies (EASTS) Conference, 2025.

Coordinator: Overall Coordinator, Annual Conference on Infrastructure (IBSR), IIT Kharagpur (2023 & 2024).

Reviewer: Transport Policy, Transportation Research Part A & D, RTBM, TRR (SAGE), Transportation in Developing Economies.

Memberships: ASCE T&DI, TRG India, WCTR Society, IATBR.

Awards

- **Best Presentation Award**, Research Scholar Day, IIT Kharagpur.
- **Best Poster Presentation**, Cyber-Physical System Summit (CyPhySS 2023).
- **Travel Grant**, University of Leeds (Research Exchange).
- **GATE Qualified** (Rank 559), Civil Engineering, 2019.
- **MHRD Fellowship** for PhD Research.
- Winner at Regional level in National Children Science Congress (DST, Govt of India).

Referees

Prof. Arkopal Kishore Goswami

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